

GOKULRAJ E

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PROFESSIONAL SUMMARY

Backend developer specializing in Python and FastAPI, with a local, privacy-preserving Retrieval-Augmented Generation (RAG) system built end-to-end — from custom retrieval filtering to containerized deployment — and a peer-reviewed publication on the same work. Comfortable owning a backend service from database schema through API to deployment without hand-holding.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, Java

Backend: FastAPI, Django, REST API Development, Authentication, CRUD Operations

Databases: MySQL, SQLite, Qdrant (vector database)

AI/ML Technologies: LangChain, Ollama, BAAI/bge-m3 embeddings, Retrieval-Augmented Generation (RAG), OCR (Apache Tika, Tesseract)

DevOps & Tools: Docker, Git, GitHub, Postman, VS Code

Software Engineering Concepts: Object-Oriented Programming (OOP), Database Design, Version Control, API Development

Frontend: HTML, CSS, JavaScript, PHP

PROJECTS

Privacy-Preserving RAG (Retrieval-Augmented Generation) System

Jan 2026

Local, privacy-preserving backend enabling secure document retrieval and question-answering with no external API calls, ensuring complete data sovereignty for sensitive enterprise documents.

- Engineered a backend RAG pipeline with FastAPI and Ollama, removing all external API dependencies to achieve 100% on-premise data processing.
- Designed a high-precision retrieval layer using BAAI/bge-m3 embeddings with a custom gap-based relevance filtering algorithm to reduce semantic noise and hallucinations across multi-turn queries.
- Integrated a dual-engine OCR pipeline (Apache Tika and Tesseract) to automate text extraction from diverse, unstructured enterprise file formats.
- Architected and managed a Qdrant vector database for semantic search and document indexing.
- Containerized the full application stack with Docker for consistent, reproducible local deployment.

Tech Stack: Python, FastAPI, LangChain, Qdrant, Ollama, Docker

Uzhavar Koodam – Agricultural Market Platform

Responsive web platform giving farmers real-time agricultural market prices to support informed, data-driven selling decisions.

- Developed the backend with PHP and MySQL to store and serve real-time market price data.
- Built an interactive crop selection feature with filtering and autocomplete, cutting search time by ~30%.
- Designed a responsive frontend with HTML, CSS, and JavaScript for cross-device accessibility.
- Structured the database schema to support efficient querying of crop and price records.

Tech Stack: HTML, CSS, JavaScript, PHP, MySQL

Real-Time Weather Forecast Application

- Integrated a third-party REST weather API to fetch and display real-time forecast data.
- Implemented client-side state management to handle dynamic weather data updates.
- Designed a responsive, component-based UI with HTML, CSS, and JavaScript.

Tech Stack: JavaScript, HTML, CSS, REST API Integration

EDUCATION

B.Tech in Information Technology

Kalasalingam Academy of Research and Education, Virudhunagar, Tamil Nadu | 2022 – 2026

CGPA: 6.99 / 10.0

PUBLICATIONS

“Privacy-Preserving RAG: A Local System for Enterprise Document Retrieval” — Presented at the 4th International Conference on Inventive Computing and Informatics (ICICI 2026), 10–12 June 2026, S.E.A. College of Engineering and Technology, Bengaluru, Karnataka, India.

LANGUAGES

English, Tamil